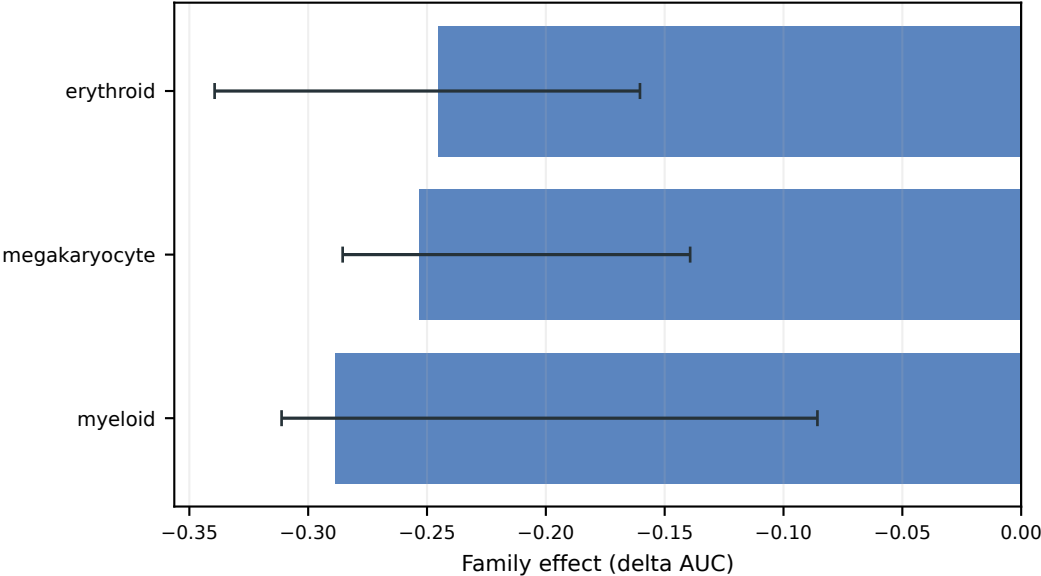


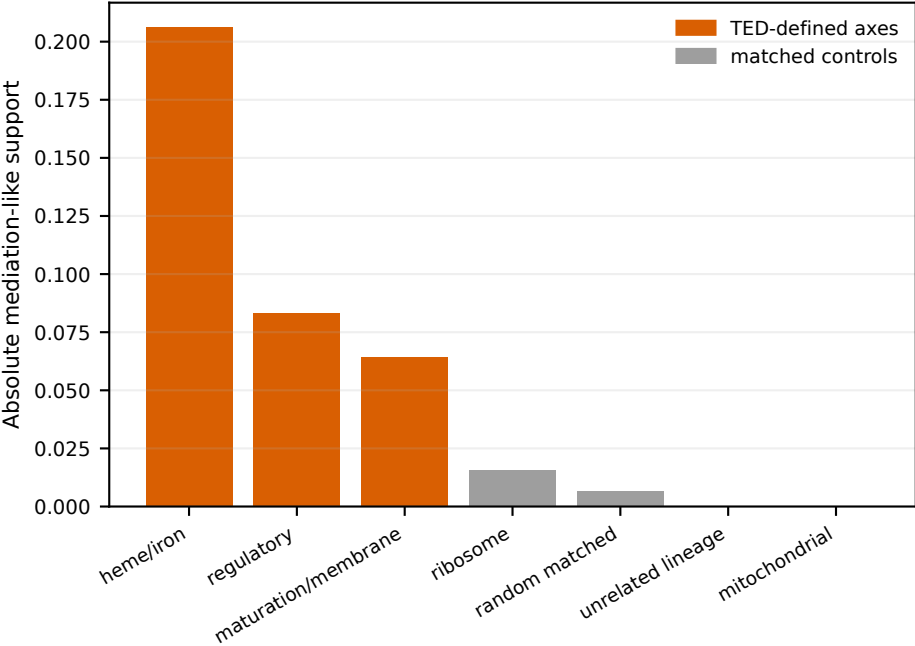
GSE271399 erythroid event-loss with independent GATA1/GATA1s directional support

A. Primary GSE271399 event-loss family

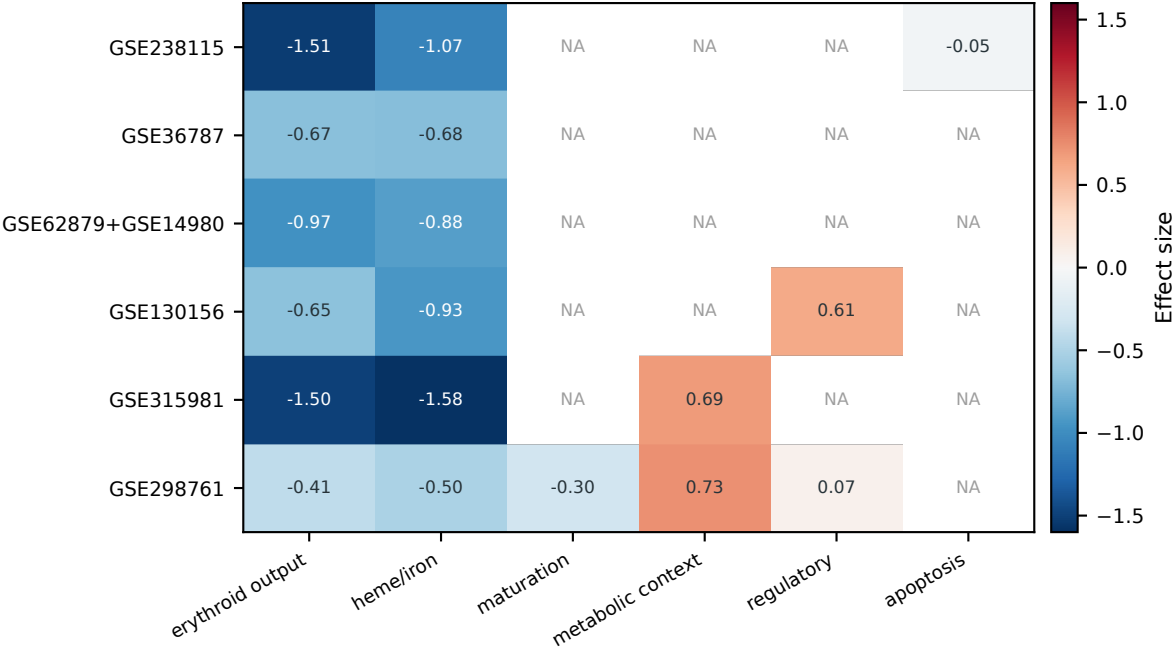
$q_{block} = 0.0008$; direction stability = 1.0



B. Erythroid/heme axes exceed matched controls



C. Independent GATA1/GATA1s datasets support direction



D. Boundary gates: support added, rescue missing

PASS	Primary event robust
PASS	Independent direction consistency = 1.0
PASS	No global negative-control failure
PASS	Cross-dataset supported event
MISSING	Matched rescue design
MISSING	Same-system rescue evidence
MISSING	Functional-causal rescue boundary

Boundary retained: cross-dataset-supported computational candidate
Blocked escalation: matched functional rescue / causal validation